Extracted from: A Taste of Topology, Volker Runde

3

Set-Theoretic Topology

Why would one want to attempt to extend notions such as convergence and continuity to a setting even more abstract than metric spaces?

The answer is that, already at a very elementary level, one encounters phenomena that do not fit into the framework of metric spaces: pointwise convergence, for instance—the most basic notion of convergence there is for functions—cannot be described as convergence with respect to a metric (as we show in this chapter).

Convergence and continuity in the metric setting were based on a notion of "closeness" for points: two points were sufficiently close if their distance, measured through the given metric, was sufficiently small. Going beyond metric spaces and still being able to meaningfully speak of convergence and continuity therefore ought to be based on an axiomatized notion of closeness. Such an axiomatization exists (and is surprisingly simple): it lies at the heart of the concept of a topological space. (For technical reasons, we pursue a slightly different, but equivalent route.)

3.1 Topological Spaces—Definitions and Examples

A topological space is supposed to be a set that has just enough structure to meaningfully speak of continuous functions on it. In view of Corollary 2.3.10, a reasonable approach would be to axiomatize the notion of an open set:

Definition 3.1.1. *Let X be a set. A* topology *on X is a subset $\mathcal{T}$ of $\mathfrak{P}(X)$ such that:*

(a) $\varnothing, X \in \mathcal{T}$;
(b) *If $\mathcal{U} \subset \mathcal{T}$ is arbitrary, then $\bigcup\{U : U \in \mathcal{U}\}$ lies in $\mathcal{T}$;*
(c) *If $U_1, U_2 \in \mathcal{T}$, then $U_1 \cap U_2 \in \mathcal{T}$.*

denoting the set of all subsets of X

The sets in $\mathcal{T}$ are called open. *A set together with a topology is called a* topological space.

We often write $(X, \mathcal{T})$ for a topological space X with topology $\mathcal{T}$; sometimes, if the topology is obvious or irrelevant, we may also simply write X.

Examples 3.1.2. (a) Let (X, d) be a metric space, and let $\mathcal{T}$ denote the collection of all subsets of X that are open in the sense of Definition 2.2.3. By Proposition 2.2.5, $\mathcal{T}$ is indeed a topology. It is clear that $\mathcal{T}$ does not depend on the particular metric d, but only on its equivalence class: any metric on X equivalent to d yields the same topology. Topological spaces of this type are called *metrizable*.

(b) Let X be any set, and let $\mathcal{T} = \mathfrak{P}(X)$. This is just a special case of the first example: equip X with the discrete metric. Such topological spaces are called *discrete*.

(c) Let X be any set, and let $\mathcal{T} = \{\varnothing, X\}$. Such topological spaces are called *chaotic*.

(d) Let X be any set, and let $\mathcal{T}$ consist of $\varnothing$ and all subsets of X with finite complement.

(e) Let X be any set, and let $\mathcal{T}$ consist of $\varnothing$ and all subsets of X with countable complement.

(f) Let $(X, \mathcal{T})$ be a topological space, and let $Y \subset X$. The *relative topology* on Y (or the *topology inherited from* X) is the collection

$$\mathcal{T}|_Y := \{Y \cap U : U \in \mathcal{T}\}$$

of subsets of Y. It is clearly a topology on Y. The space $(Y, \mathcal{T}|_Y)$ is then called a *subspace* of X.

Is every topological space metrizable? Of course not, and here is why.

Definition 3.1.3. *A topological space $(X, \mathcal{T})$ is called* Hausdorff *if, for any $x, y \in X$ with $x \neq y$, there are sets $U, V \in \mathcal{T}$ with $x \in U$, $y \in V$, and $U \cap V = \varnothing$.*

Informally, Definition 3.1.3 is often expressed as, "In a Hausdorff space, points can be separated by open sets."

Examples 3.1.4. (a) Let (X, d) be a metric space, and let $x, y \in X$ be such that $x \neq y$. It follows that $\epsilon := \frac{1}{2}d(x, y) > 0$. Let $U := B_\epsilon(x)$ and let $V := B_\epsilon(y)$. It follows that $U \cap V = \varnothing$, so that X is Hausdorff.

(b) If X is any set with more than one element, then X equipped with the chaotic topology is not Hausdorff (and therefore not metrizable).

(c) Let X be an infinite set equipped with the topology from Example 3.1.2(d), and let $x, y \in X$ be such that $x \neq y$. Assume that X is Hausdorff. Then there are open sets U and V of X with $x \in U$, $y \in V$, and $U \cap V = \varnothing$. This, however, entails that $X = (X \setminus U) \cup (X \setminus V)$ is finite, which is a contradiction.

(d) Similarly, if X is any uncountable set equipped with the topology from Example 3.1.2(e), the resulting topological space fails to be Hausdorff (see Exercise 1 below).

We soon encounter Hausdorff spaces that nevertheless fail to be metrizable.

With a notion of open sets at hand, we can define, of course, what a closed subset of a topological space is supposed to be:

Definition 3.1.5. *Let $(X, \mathcal{T})$ be a topological space. A subset F of X is called* closed *if $X \setminus F$ is open.*

As for metric spaces, we have the following.

Proposition 3.1.6. *Let $(X, \mathcal{T})$ be a topological space. Then:*

(i) *$\varnothing$ and X are closed.*
(ii) *If $\mathcal{F}$ is a family of closed subsets of X, then $\bigcap\{F : F \in \mathcal{F}\}$ is closed.*
(iii) *If F_1 and F_2 are closed subsets of X, then $F_1 \cup F_2$ is closed.*

Of course, one can also define a topology on a given set by declaring certain sets as closed, then checking that these sets satisfy Proposition 3.1.6(i) through (iii), and defining their complements as open; it is clear that this approach is equivalent to Definition 3.1.1.